

# Evaluating Expressions with Radicals

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**MA.8.NSO.1.7**

*I can evaluate an expression that contains square roots, cube roots and integer exponents.*

*Yesterday every number was a measurement.*

**Today the radical sign comes back — and its bar is a grouping symbol.**

# Warm-Up

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*On your own. Four minutes. No notes.*

1. How many significant digits are in 0.0250 ?
2.  $(3 \times 10^4)(2 \times 10^{-7})$
3. Find the value:  $\sqrt{196}$
4. Evaluate:  $20 - 3 \cdot 4$

# Warm-Up

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*Answers.*

1. How many significant digits are in 0.0250 ? **3**
2.  $(3 \times 10^4)(2 \times 10^{-7})$   **$6 \times 10^{-3}$**
3. Find the value:  $\sqrt{196}$  **14**
4. Evaluate:  $20 - 3 \cdot 4$  **8**

# Notes

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*Copy all three lines.*

## **I. The radical bar is a grouping symbol**

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$$\sqrt{25} - \sqrt{36} + \sqrt{1} = 5 - 6 + 1 = 0$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4$$

$$\sqrt{9 + 16} = \sqrt{25} = 5 \quad , \quad \text{never} \quad 3 + 4$$

**Everything under the bar is evaluated before the root is taken.**

# Notes

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## II. Square roots and cube roots are not the same shape

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$$\sqrt{49} = 7$$

$$\sqrt[3]{64} = 4$$

$$\sqrt[3]{-64} = -4$$

$$\sqrt{\frac{1}{9}} = \frac{1}{3}$$

$$\sqrt[3]{\frac{1}{125}} = \frac{1}{5}$$

**An odd root keeps the sign of its radicand. A square root sign never gives a negative value.**

# Notes

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## III. Radicals alongside exponents

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$$5^{-3} = \frac{1}{5^3} = \frac{1}{125}$$

$$\sqrt{4^3 - 39} = \sqrt{25} = 5$$

$$\sqrt{4^3 - 39} \cdot 5^{-3} = 5 \cdot \frac{1}{125} = \frac{1}{25}$$

**Finish each piece — the radical, the power — and only then put them together.**

# Example 1

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*A power and a radical in one expression.*

$$\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$$

**Evaluate.**

# Example 1

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*Worked.*

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4} \quad \text{square the whole base, sign included}$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4 \quad \text{finish under the bar, then take the root}$$

$$\frac{1}{4} + 4 = \frac{17}{4} \quad \text{combine}$$

**Answer: 17/4**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

# Your Turn

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*Answer.*

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

1/9 minus 2

**Answer:  $-17/9$**

# Example 2

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*A radical inside a product.*

$$\sqrt[3]{27} - 1.4(\sqrt{3^2 - 5})$$

**Evaluate.**

# Example 2

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*Worked.*

$$\sqrt[3]{27} = 3, \quad 3^2 - 5 = 4 \quad \text{the two innermost pieces}$$

$$\sqrt{4} = 2, \quad 1.4(2) = 2.8 \quad \text{the root, then the product}$$

$$3 - 2.8 = 0.2 \quad \text{subtract last}$$

**Answer: 0.2**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

# Your Turn

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*Answer.*

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

−2 plus 10

**Answer: 8**

# Whiteboards • Question 1 of 9

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*Boards up on three.*

$$\sqrt{121}$$

**Answer it.**

*What times itself?*

# Whiteboards • Question 1 of 9

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*Answer.*

$$\sqrt{121}$$

$$11 \cdot 11$$

**Answer: 11**

# Whiteboards • Question 2 of 9

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*Boards up on three.*

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

**Answer it.**

*A square's side is the root of its area.*

# Whiteboards • Question 2 of 9

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*Answer.*

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

$$\sqrt{169}$$

**Answer: 13 feet**

# Whiteboards • Question 3 of 9

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*Boards up on three.*

$$\sqrt[3]{-125}$$

**Answer it.**

*What number, three times?*

# Whiteboards • Question 3 of 9

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*Answer.*

$$\sqrt[3]{-125}$$

$$(-5)(-5)(-5)$$

**Answer: -5**

# Whiteboards • Question 4 of 9

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*Boards up on three.*

$$\sqrt{2^3 + 17}$$

**Answer it.**

*Finish under the bar first.*

# Whiteboards • Question 4 of 9

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*Answer.*

$$\sqrt{2^3 + 17}$$

$$8 + 17 = 25$$

**Answer: 5**

# Whiteboards • Question 5 of 9

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*Boards up on three.*

$$\sqrt{9 + 16}$$

Which is the value?

**A.** 5

**B.** 7

**C.** 25

**D.** 12.5

**Answer it.**

# Whiteboards • Question 5 of 9

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*Answer.*

$$\sqrt{9 + 16}$$

Which is the value?

**A. 5**

**B. 7**

**C. 25**

**D. 12.5**

**Answer: A — 5**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

$$\sqrt[3]{-64}$$

Which is the value?

**A.** -4

**B.** 4

**C.** -8

**D.** not a real number

**Answer it.**

# Whiteboards • Question 6 of 9

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*Answer.*

$$\sqrt[3]{-64}$$

Which is the value?

**A. -4**

**B. 4**

**C. -8**

**D. not a real number**

**Answer: A — -4**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

**Answer it.**

*Top and bottom each get the root.*

# Whiteboards • Question 7 of 9

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*Answer.*

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

1/3 and 1/3

**Answer: 2/3**

# Whiteboards • Question 8 of 9

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*Boards up on three.*

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

**Answer it.**

*Finish each piece, then multiply.*

# Whiteboards • Question 8 of 9

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*Answer.*

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

5 times one hundred twenty-fifth

**Answer: 1/25**

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Simon says that  $\sqrt[3]{-27}$  is not a real number,

because you cannot take the root of a negative.

Explain why he is wrong, and give the value.

**Write your answer in sentences.**

*Which root is he thinking of?*

# Whiteboards • Question 9 of 9

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*Answer.*

Simon says that  $\sqrt[3]{-27}$  is not a real number,

because you cannot take the root of a negative.

Explain why he is wrong, and give the value.

a cube root is not a square root

**Answer: -3**

# Independent Set

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*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Evaluate radical expressions — D8J